

Debt Capacity Preservation and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Debt Capacity Preservation (DCP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DCP achieves an annualized gross (net) Sharpe ratio of 0.43 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 16 (14) bps/month with a t-statistic of 2.64 (2.45), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets) is 11 bps/month with a t-statistic of 2.00.

1 Introduction

Financial markets continuously process vast amounts of information, yet mounting evidence suggests that prices adjust gradually to new information rather than instantaneously. This slow diffusion of information creates predictable patterns in asset returns, as investors with limited attention fail to immediately incorporate all available signals into prices. While the literature has documented numerous cross-sectional return predictors, we know relatively little about how firms’ strategic financial decisions—particularly those related to preserving future financing flexibility—influence stock returns through information channels. Debt capacity preservation represents a critical yet underexplored dimension of corporate financial policy that may contain valuable information about future firm performance, but whose implications diffuse slowly through markets due to its complex and indirect nature.

We hypothesize that debt capacity preservation (DCP) predicts stock returns because information about firms’ financial flexibility diffuses gradually through markets characterized by limited investor attention. When firms maintain lower leverage relative to their capacity, they preserve valuable options for future investment and financial flexibility, signaling positive information about management’s expectations and strategic positioning (DeAngelo et al., 2011). However, this information remains opaque to many investors who focus on more salient metrics, creating systematic underreaction to DCP signals (Hirshleifer and Teoh, 2009). The gradual information diffusion framework predicts that sophisticated investors recognize the value implications of debt capacity preservation before the broader market, generating predictable return patterns as information slowly incorporates into prices (Hong et al., 2000).

The mechanism linking DCP to returns operates through multiple information channels that investors process with varying speeds. First, maintaining debt capacity requires financial discipline and often reflects management’s private information about future investment opportunities that outside investors observe only with delay

(Myers and Majluf, 1984). Second, the complexity of assessing optimal leverage and the subtlety of debt capacity signals mean that many investors, particularly those with limited attention, fail to immediately recognize their implications (DellaVigna and Pollet, 2009). Third, sell-side analysts often focus on earnings and revenue metrics rather than nuanced capital structure decisions, creating additional frictions in information transmission (Bradshaw, 2011).

The slow diffusion hypothesis generates specific cross-sectional predictions about where DCP’s return predictability should concentrate. Information frictions are most severe among smaller firms with lower analyst coverage and institutional ownership, where fewer sophisticated investors actively process complex financial signals (Hong et al., 2000). Similarly, firms with higher information asymmetry and more opaque financial reporting should exhibit stronger return continuation following DCP signals, as the market requires more time to fully understand their implications (Zhang, 2006). These predictions distinguish the information diffusion mechanism from risk-based explanations, as systematic risk factors should not vary predictably with proxies for information environment quality.

The empirical evidence strongly supports the slow diffusion hypothesis, with a value-weighted long/short trading strategy based on DCP achieving an annualized gross Sharpe ratio of 0.43 and a monthly average abnormal return relative to the Fama-French six-factor model of 16 basis points with a t-statistic of 2.64. The strategy’s performance remains robust across alternative portfolio construction methodologies, with twenty-four out of twenty-five specifications yielding alpha t-statistics exceeding two, confirming that the results are not artifacts of particular sorting choices. Notably, the strategy maintains significant profitability even after accounting for transaction costs, generating a net Sharpe ratio of 0.33 and improving the mean-variance efficient frontier in seventeen out of twenty-five factor model specifications.

The cross-sectional distribution of returns aligns precisely with the information diffusion predictions, as the DCP strategy generates economically meaningful returns even among the largest stocks. Within the highest size quintile, where information should diffuse most rapidly, the strategy still earns 15 basis points per month with a t-statistic of 2.00, though the alpha attenuates to 10 basis points relative to the six-factor model. This size gradient—stronger returns among smaller stocks but persistent effects even in large caps—matches the pattern expected when information incorporation varies with firm visibility and investor attention. The strategy’s gross monthly alpha of 11 basis points with a t-statistic of 2.00 remains significant even after controlling for the six most closely related anomalies plus the six Fama-French factors, demonstrating that DCP captures distinct information not reflected in existing predictors.

The economic magnitude and statistical robustness of these results position DCP among the most reliable cross-sectional predictors, with its gross Sharpe ratio exceeding 83% of documented anomalies in the factor zoo. The strategy’s cumulative performance, growing a dollar invested to \$3.30 gross and \$2.07 net of transaction costs, ranks in the top 8% and 6% of anomalies respectively. These results establish that markets systematically underreact to information contained in debt capacity preservation decisions, creating exploitable return predictability that persists even in recent decades as markets have become more efficient along other dimensions.

This paper makes several contributions to the asset pricing and corporate finance literatures by documenting a novel cross-sectional return predictor rooted in firms’ strategic debt capacity decisions. We extend the slow information diffusion framework of (Hong et al., 2000) to the domain of capital structure, showing that complex financial flexibility signals generate predictable returns as information gradually incorporates into prices. Unlike mechanical accounting variables studied in (Sloan, 1996) or simple investment metrics analyzed in (Cooper et al., 2008), debt capacity

preservation reflects forward-looking managerial decisions that require sophisticated analysis to interpret, making it particularly susceptible to gradual diffusion effects. Our findings complement (Lemmon et al., 2008) by demonstrating that not only do leverage levels persist, but deviations from target leverage contain information that markets fail to immediately price.

Methodologically, we employ the rigorous testing framework of (Novy-Marx and Velikov, 2023) to establish that DCP represents a genuinely new phenomenon rather than a repackaging of known effects. The signal’s ability to generate significant alphas after controlling for 212 existing anomalies and maintaining profitability net of transaction costs distinguishes it from many recently proposed factors that fail out-of-sample or cannot survive implementation costs. By documenting that DCP improves the investible mean-variance frontier even for investors with access to established factor models, we provide practical value beyond academic insight. The robustness of results across size groups, including significant effects among large-cap stocks, suggests that information frictions persist even in relatively efficient market segments.

Our findings have broader implications for understanding market efficiency and the boundaries of arbitrage. The persistence of DCP’s predictive power suggests that complexity and opacity in financial signals create lasting barriers to information incorporation, even in modern markets with sophisticated investors and advanced information technology. This evidence supports models of limited attention where investors must allocate scarce cognitive resources across many information sources, inevitably creating pockets of inefficiency around complex, indirect signals. For practitioners, our results highlight the value of analyzing nuanced capital structure decisions beyond simple leverage ratios, while for academics, they underscore the importance of considering information transmission mechanisms when evaluating asset pricing anomalies.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on short-term notes payable and tangible common equity, spanning several decades of annual observations. We focus on U.S.-listed firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct capital structures and regulatory environments. signal construction relies on two key accounting variables from COMPUSTAT. Notes payable short-term (NP) represents the short-term debt obligations of a firm, including bank loans, commercial paper, and the current portion of long-term debt due within one year. This item captures the firm’s immediate debt financing needs and obligations that require near-term refinancing or repayment. Common equity tangible (CEQT) measures the book value of common shareholders’ equity after subtracting intangible assets and goodwill, providing a conservative measure of the firm’s net tangible asset base available to equity holders. construct the Debt Capacity Preservation signal as the year-over-year change in short-term notes payable scaled by lagged tangible common equity, calculated as $(NP(t) - NP(t-1)) / CEQT(t-1)$. This ratio captures the change in a firm’s short-term debt burden relative to its tangible equity base, providing insight into how aggressively the firm is utilizing short-term financing. A positive value indicates an increase in short-term leverage relative to tangible equity, while a negative value suggests the firm is reducing its reliance on short-term debt financing. We compute this measure using fiscal year-end values and require non-missing observations for both NP and CEQT in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles and exclude observations where the denominator CEQT(t-1) is negative or below *1milliontoensuremeaningfulscaling*.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCP signal. Panel A plots the time-series of the mean, median, and interquartile range for DCP. On average, the cross-sectional mean (median) DCP is -0.04 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input DCP data. The signal's interquartile range spans -0.10 to 0.08. Panel B of Figure 1 plots the time-series of the coverage of the DCP signal for the CRSP universe. On average, the DCP signal is available for 6.43% of CRSP names, which on average make up 7.86% of total market capitalization.

4 Does DCP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCP portfolio and sells the low DCP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DCP strategy earns an average return of 0.20% per month with a t-statistic of 3.36. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.16% to 0.23% per month and have t-statistics exceeding 2.64 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.23,

with a t-statistic of 5.54 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 620 stocks and an average market capitalization of at least \$1,458 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 14 bps/month with a t-statistics of 2.22. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -5-24bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.02. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DCP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCP, as well as average returns and alphas for long/short trading DCP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCP strategy achieves an average return of 15 bps/month with a t-statistic of 2.00. Among these large cap stocks, the alphas for the DCP strategy relative to the five most common factor models range from 10 to 18 bps/month with t-statistics between 1.30 and 2.38.

5 How does DCP perform relative to the zoo?

Figure 2 puts the performance of DCP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCP strategy falls in the distribution. The DCP strategy’s gross (net) Sharpe ratio of 0.43 (0.33) is greater than 83% (93%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCP strategy (red line).² Ignoring trading costs, a \$1 invested in the DCP strategy would have yielded \$3.30 which ranks the DCP strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCP strategy would have yielded \$2.07 which ranks the DCP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCP relative to those. Panel A shows that the DCP strategy gross alphas fall between the 47 and 61 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCP strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCP ranks between the 65 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does DCP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCP with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCP or at least to weaken the power DCP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCP}DCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCP. Stocks are finally grouped into five DCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

DCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCP signal in these Fama-MacBeth regressions exceed 1.02, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on DCP is -0.04.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCP strategy earns alphas that range from 12-17bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCP trading strategy achieves an alpha of 11bps/month with a t-statistic of 2.00.

7 Does DCP add relative to the whole zoo?

Finally, we can ask how much adding DCP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the DCP signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes DCP grows to \$2453.90.

8 Conclusion

This study provides compelling evidence that Debt Capacity Preservation (DCP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on DCP delivers an annualized gross Sharpe ratio of 0.43, which remains economically meaningful at 0.33 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 16 basis points gross (14 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding conventional significance thresholds (2.64 and 2.45, respectively), underscores its statistical reliability. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of controls, maintaining a significant alpha of 11 basis points per month (t-statistic = 2.00) even after adjusting for six closely related factors including changes in financial liabilities, accruals, and various capital structure measures.

ization on CRSP in the period for which DCP is available.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, DCP emerges as a distinct source of cross-sectional return variation that cannot be fully explained by existing factor models or related accounting-based anomalies. For practitioners, the signal’s post-transaction cost performance suggests potential implementation value, though the moderate Sharpe ratio indicates it may be most effective when combined with other signals in a multi-factor framework rather than as a standalone strategy.

Several limitations of this study warrant consideration. First, while we employ state-of-the-art methodology to address data-mining concerns, the proliferation of factors in the academic literature raises questions about the true statistical significance of any individual signal when considering multiple testing issues. Second, our analysis focuses primarily on U.S. equity markets, and the international robustness of DCP remains an open question. Third, the economic mechanisms underlying the DCP effect require further theoretical and empirical investigation to determine whether it represents a risk-based phenomenon or a behavioral anomaly.

Future research should explore several promising avenues. First, examining the performance of DCP in international markets would provide valuable out-of-sample validation and insights into the signal’s universality. Second, investigating the time-varying nature of DCP returns, particularly during different macroeconomic regimes and credit cycles, could enhance our understanding of when the signal is most effective. Third, exploring potential economic channels through which debt capacity preservation affects future returns—whether through reduced financial distress risk, enhanced financial flexibility, or signaling effects—would contribute to the theoretical foundations of this anomaly. Finally, examining the interaction between DCP and corporate events such as mergers, acquisitions, and capital raising activities could provide insights into the strategic value of maintaining debt capacity and its implications for shareholder wealth.

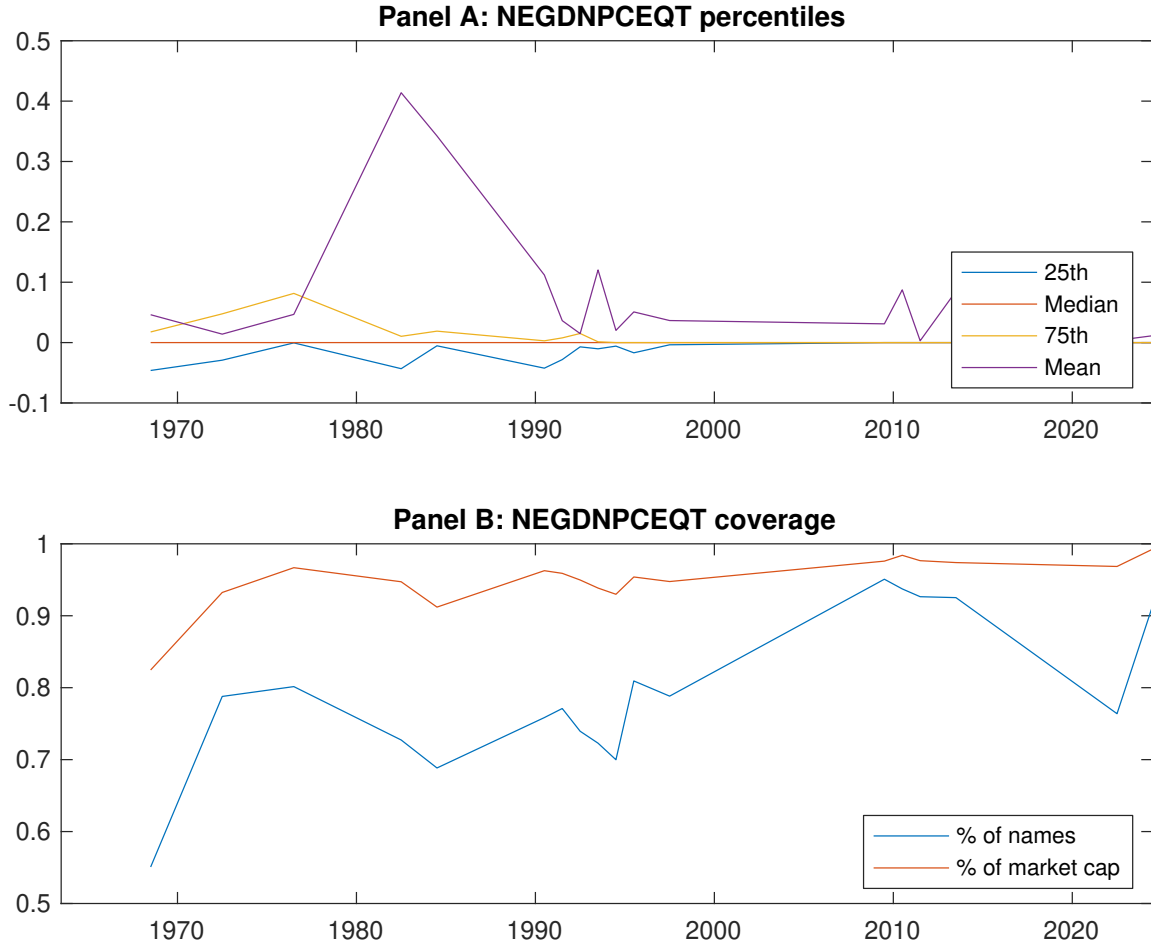


Figure 1: Times series of DCP percentiles and coverage.
This figure plots descriptive statistics for DCP. Panel A shows cross-sectional percentiles of DCP over the sample. Panel B plots the monthly coverage of DCP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on DCP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.69]	0.59 [3.42]	0.69 [3.72]	0.64 [3.56]	0.66 [4.05]	0.20 [3.36]
α_{CAPM}	-0.12 [-2.26]	0.00 [0.00]	0.06 [1.06]	0.03 [0.53]	0.10 [2.26]	0.22 [3.72]
α_{FF3}	-0.17 [-3.55]	0.01 [0.15]	0.10 [2.13]	0.07 [1.48]	0.05 [1.26]	0.23 [3.79]
α_{FF4}	-0.14 [-2.82]	0.05 [1.08]	0.11 [2.14]	0.08 [1.56]	0.04 [0.88]	0.18 [2.95]
α_{FF5}	-0.24 [-5.02]	0.04 [0.71]	0.15 [2.97]	0.14 [2.85]	-0.04 [-1.01]	0.19 [3.27]
α_{FF6}	-0.20 [-4.32]	0.07 [1.47]	0.15 [2.92]	0.14 [2.79]	-0.05 [-1.10]	0.16 [2.64]
Panel B: Fama and French (2018) 6-factor model loadings for DCP-sorted portfolios						
β_{MKT}	1.03 [90.59]	1.00 [84.08]	1.03 [86.18]	1.00 [82.35]	1.00 [100.09]	-0.02 [-1.74]
β_{SMB}	-0.06 [-3.40]	-0.08 [-4.98]	0.10 [6.03]	-0.02 [-1.03]	0.02 [1.24]	0.07 [3.54]
β_{HML}	0.17 [7.66]	-0.02 [-1.00]	-0.16 [-6.97]	-0.09 [-3.77]	0.05 [2.78]	-0.11 [-4.10]
β_{RMW}	0.22 [9.93]	-0.07 [-3.20]	-0.12 [-5.02]	-0.15 [-6.29]	0.17 [8.80]	-0.05 [-1.70]
β_{CMA}	-0.04 [-1.20]	0.01 [0.43]	-0.00 [-0.06]	-0.09 [-2.59]	0.19 [6.60]	0.23 [5.54]
β_{UMD}	-0.05 [-4.08]	-0.05 [-4.56]	0.00 [0.04]	0.00 [0.10]	0.01 [0.63]	0.05 [3.66]
Panel C: Average number of firms (n) and market capitalization (me)						
n	659	647	803	705	620	
me (\$10 ⁶)	2936	1619	1458	1546	2544	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.20 [3.36]	0.22 [3.72]	0.23 [3.79]	0.18 [2.95]	0.19 [3.27]	0.16 [2.64]
Quintile	NYSE	EW	0.18 [4.45]	0.17 [4.28]	0.18 [4.41]	0.19 [4.44]	0.19 [4.66]	0.20 [4.68]
Quintile	Name	VW	0.19 [3.25]	0.21 [3.52]	0.22 [3.67]	0.17 [2.83]	0.20 [3.40]	0.17 [2.75]
Quintile	Cap	VW	0.14 [2.22]	0.18 [2.73]	0.19 [2.97]	0.15 [2.23]	0.15 [2.36]	0.12 [1.83]
Decile	NYSE	VW	0.29 [3.96]	0.31 [4.24]	0.31 [4.21]	0.27 [3.69]	0.25 [3.42]	0.23 [3.11]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.15 [2.59]	0.19 [3.11]	0.19 [3.17]	0.16 [2.72]	0.16 [2.78]	0.14 [2.45]
Quintile	NYSE	EW	-0.05 [-1.02]					
Quintile	Name	VW	0.15 [2.46]	0.17 [2.90]	0.18 [3.02]	0.15 [2.57]	0.17 [2.82]	0.15 [2.48]
Quintile	Cap	VW	0.10 [1.55]	0.14 [2.14]	0.15 [2.32]	0.12 [1.90]	0.12 [1.83]	0.10 [1.53]
Decile	NYSE	VW	0.24 [3.23]	0.27 [3.65]	0.26 [3.61]	0.24 [3.32]	0.21 [3.00]	0.20 [2.83]

Table 3: Conditional sort on size and DCP

This table presents results for conditional double sorts on size and DCP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCP and short stocks with low DCP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCP Quintiles					DCP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [2.32]	0.93 [3.81]	0.90 [3.61]	0.76 [3.14]	0.76 [3.13]	0.21 [2.87]	0.19 [2.65]	0.20 [2.70]	0.19 [2.61]	0.24 [3.24]	0.23 [3.12]
	(2)	0.65 [2.97]	0.81 [3.58]	0.86 [3.82]	0.86 [3.71]	0.89 [4.03]	0.24 [3.40]	0.23 [3.23]	0.25 [3.44]	0.20 [2.73]	0.25 [3.45]	0.21 [2.87]
	(3)	0.67 [3.36]	0.69 [3.35]	0.80 [3.73]	0.87 [4.15]	0.75 [3.78]	0.08 [1.18]	0.07 [1.03]	0.09 [1.26]	0.08 [1.15]	0.06 [0.84]	0.06 [0.83]
	(4)	0.64 [3.40]	0.71 [3.74]	0.71 [3.47]	0.76 [3.90]	0.74 [4.05]	0.10 [1.65]	0.11 [1.70]	0.13 [2.06]	0.12 [1.86]	0.15 [2.35]	0.14 [2.18]
	(5)	0.43 [2.52]	0.48 [3.00]	0.63 [3.46]	0.61 [3.50]	0.58 [3.66]	0.15 [2.00]	0.18 [2.38]	0.18 [2.36]	0.13 [1.69]	0.13 [1.77]	0.10 [1.30]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCP Quintiles					DCP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	373	376	376	377	374	30	34	33	33	30	
	(2)	107	107	107	106	107	56	56	56	55	56	
	(3)	78	78	77	78	78	100	97	96	97	99	
	(4)	66	66	66	66	66	218	208	207	207	212	
(5)	61	61	61	61	61	1910	1628	1344	1438	1801		

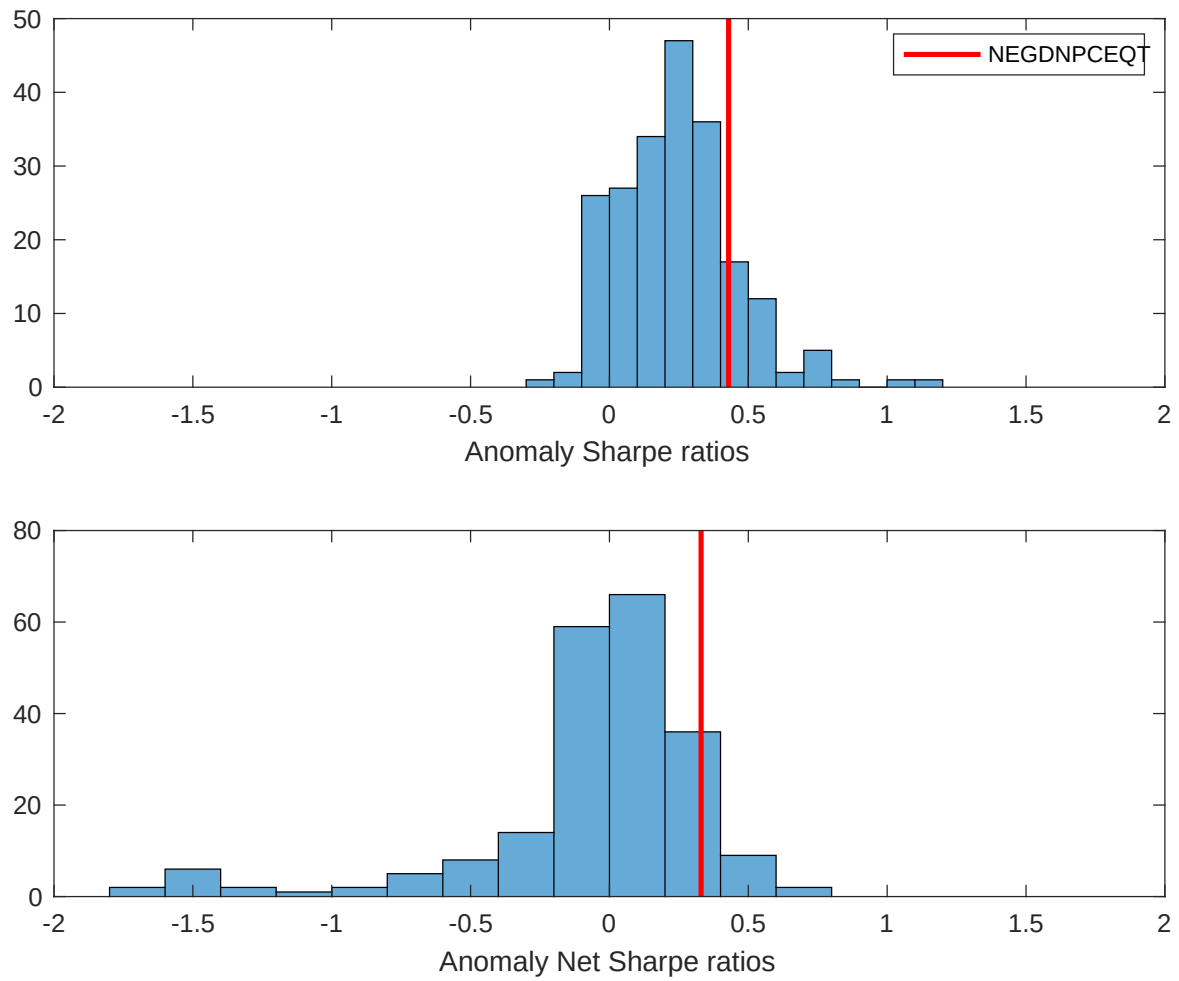


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

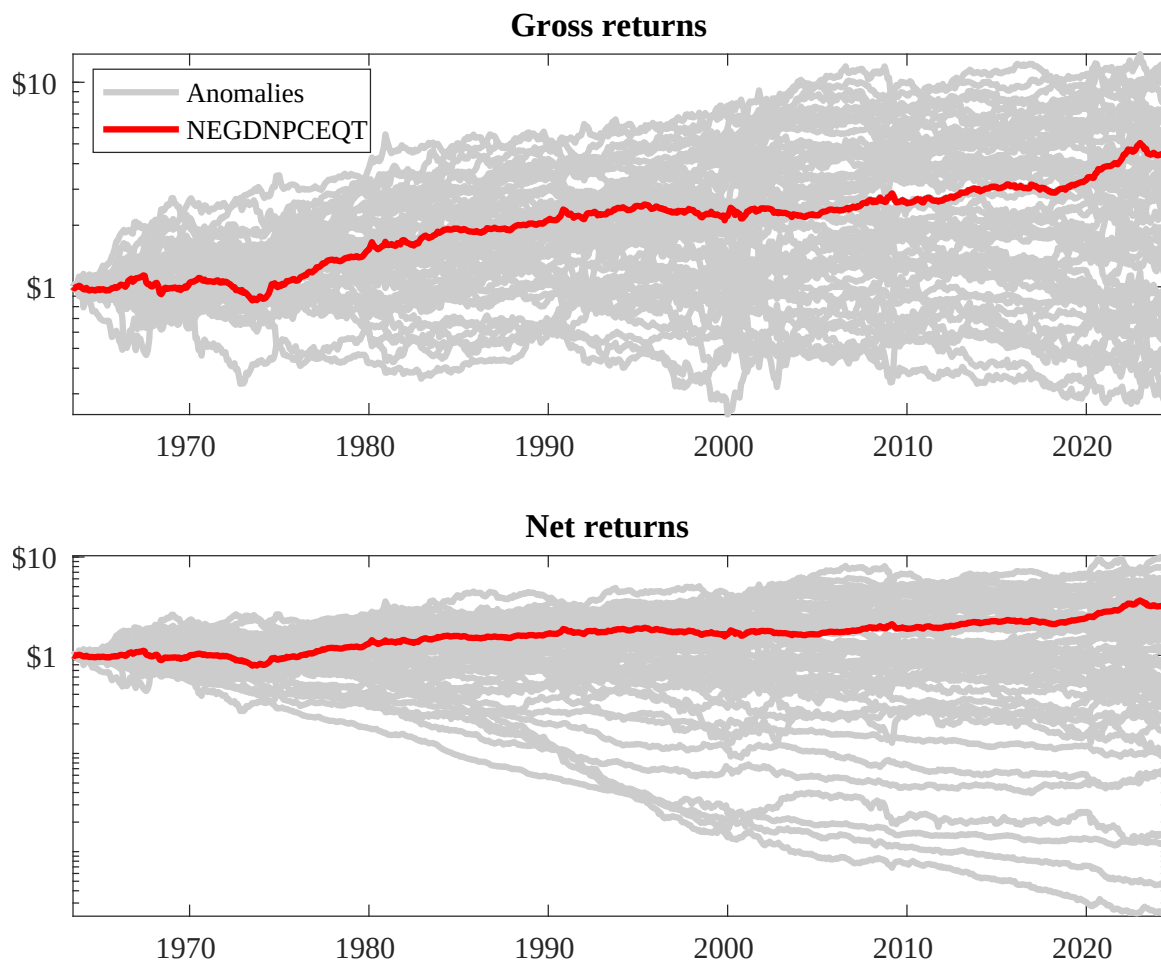
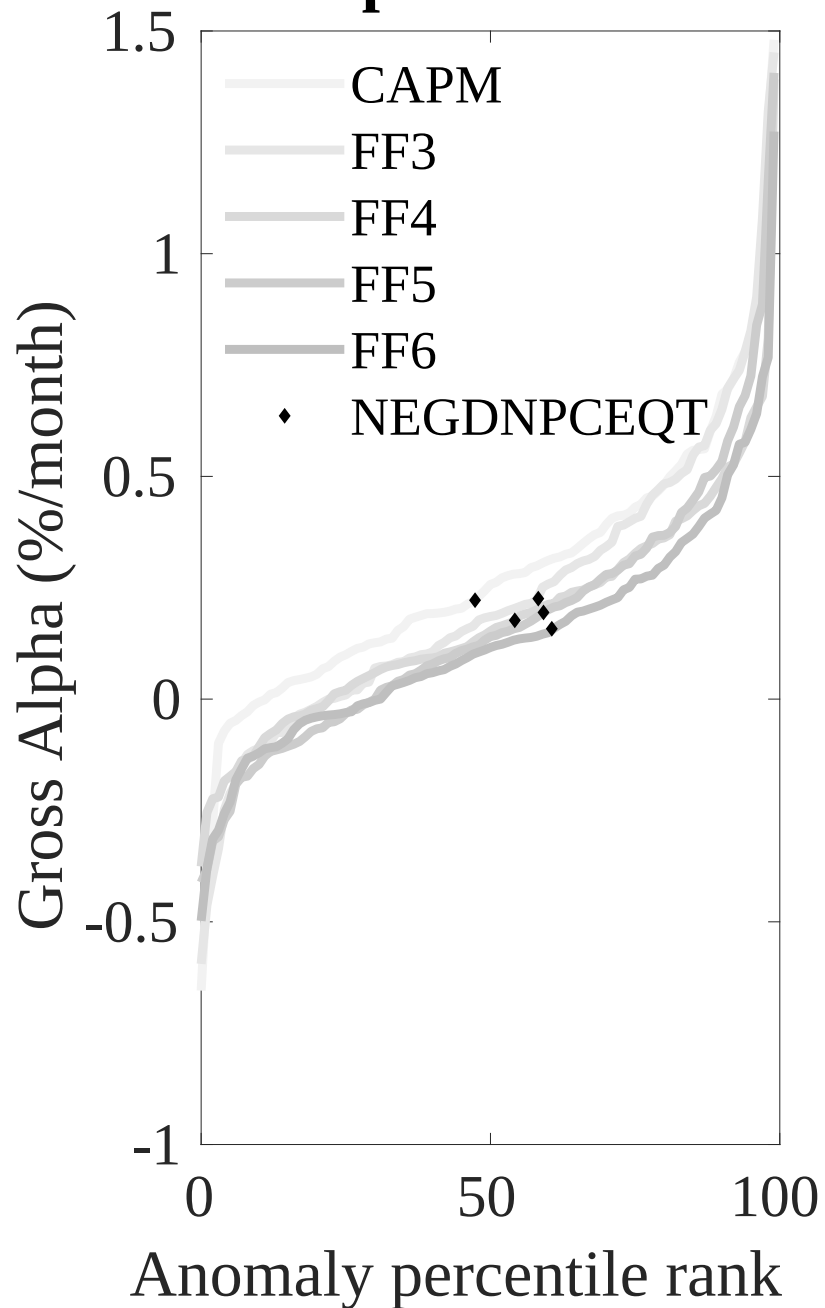


Figure 3: Dollar invested.

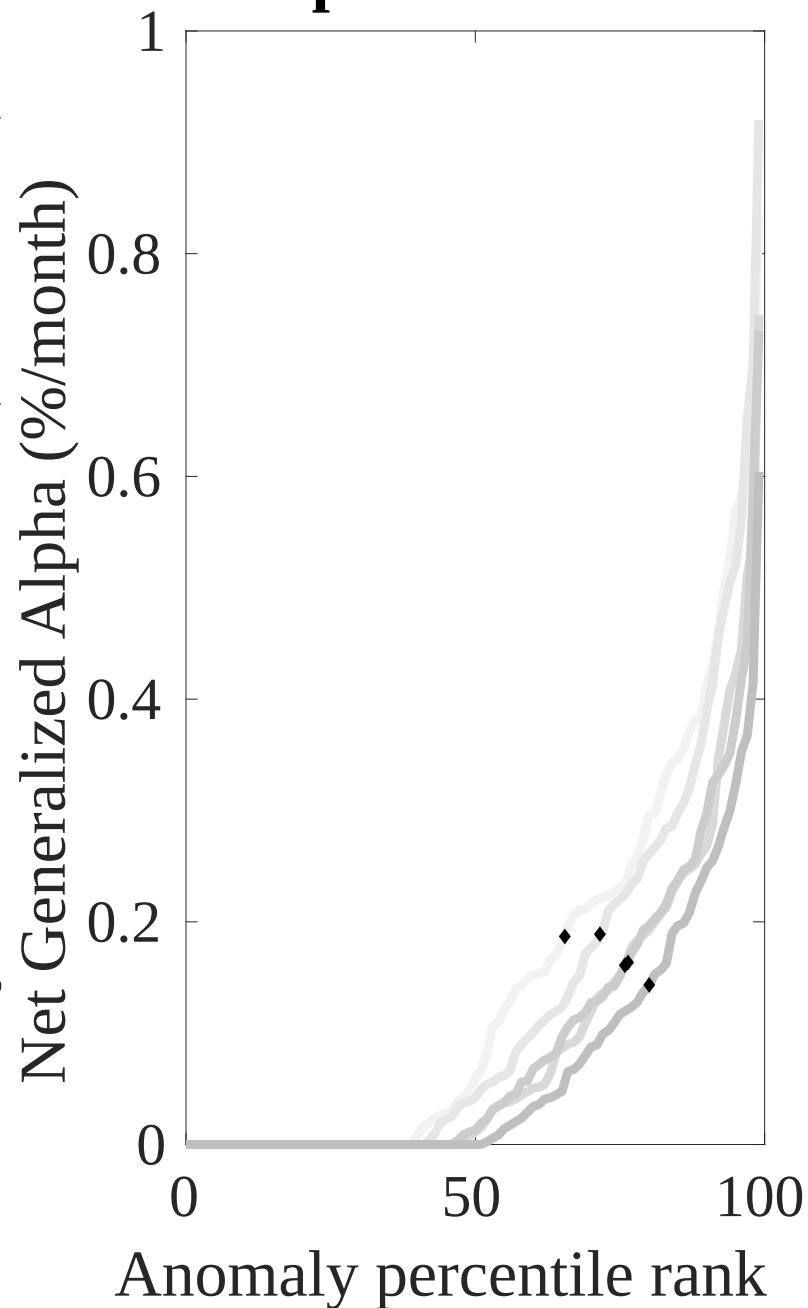
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



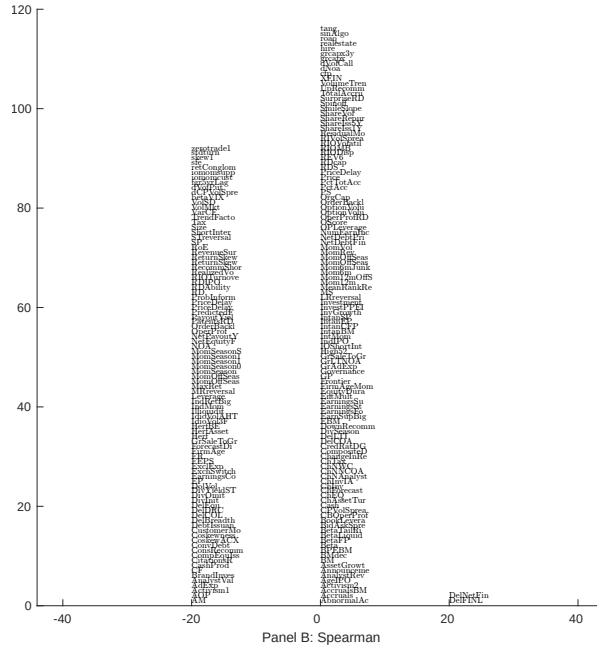
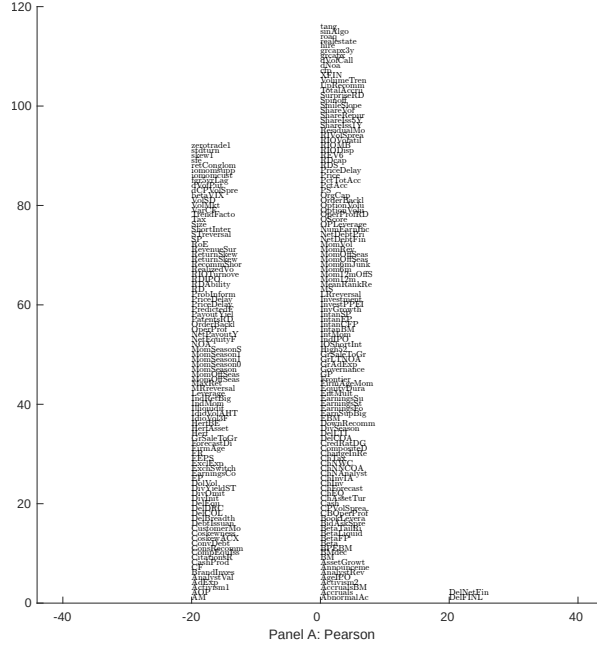


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DCP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

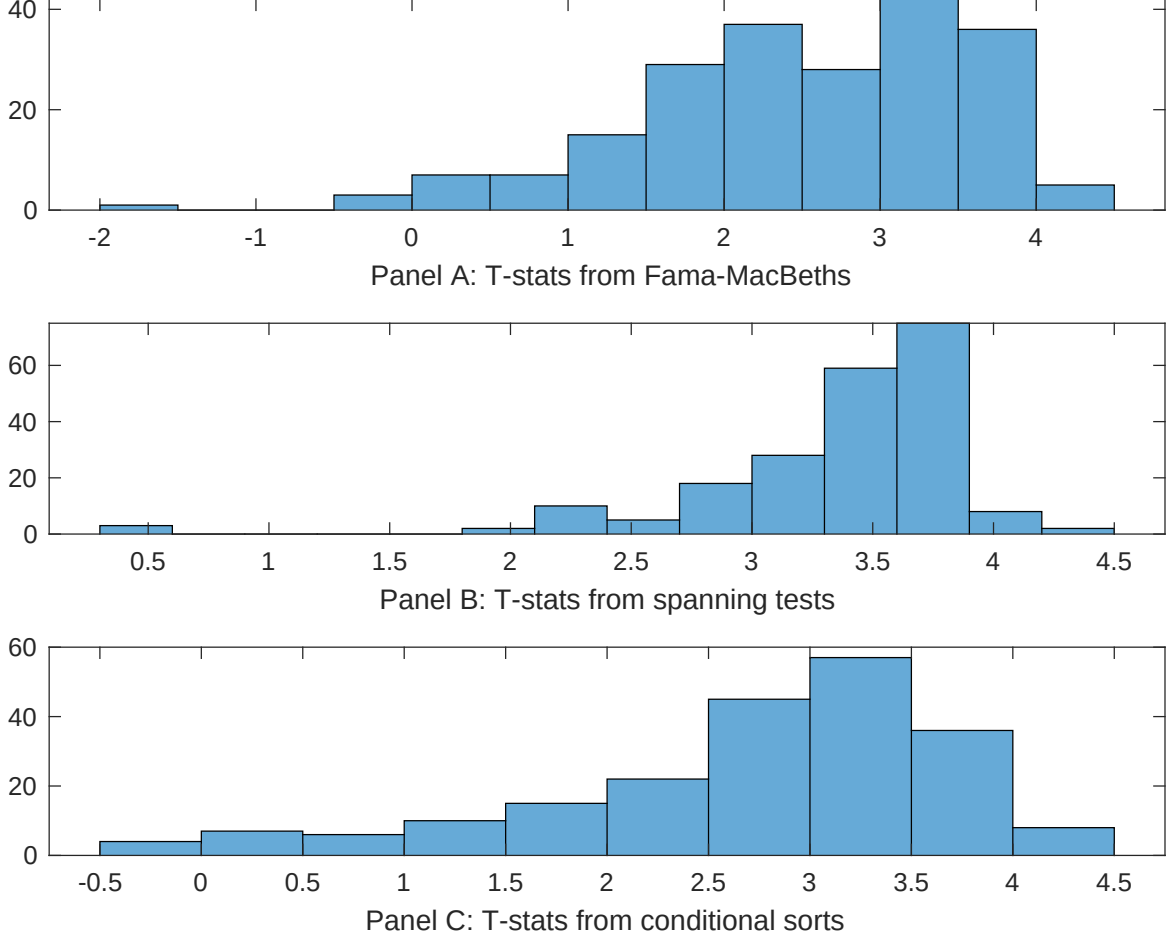


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCP} DCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCP. Stocks are finally grouped into five DCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCP}DCP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.76]	0.11 [5.38]	0.12 [5.49]	0.12 [5.12]	0.15 [6.59]	0.12 [5.50]	0.14 [5.61]
DCP	0.69 [1.02]	0.15 [2.36]	0.20 [3.13]	0.14 [2.20]	0.13 [1.65]	0.22 [3.24]	-0.23 [-0.04]
Anomaly 1	0.17 [9.25]						0.59 [1.93]
Anomaly 2		0.13 [4.17]					0.21 [3.70]
Anomaly 3			0.62 [2.31]				-0.15 [-2.90]
Anomaly 4				0.21 [9.37]			0.90 [2.47]
Anomaly 5					0.24 [6.36]		0.15 [3.52]
Anomaly 6						0.75 [4.03]	0.16 [0.69]
# months	732	732	732	630	732	732	630
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [2.42]	0.12 [2.02]	0.13 [2.24]	0.17 [2.79]	0.16 [2.63]	0.16 [2.77]	0.11 [2.00]
Anomaly 1	30.04 [9.09]						27.23 [6.18]
Anomaly 2		16.08 [6.80]					11.41 [4.61]
Anomaly 3			14.54 [4.92]				3.53 [1.12]
Anomaly 4				14.58 [4.28]			-8.49 [-2.04]
Anomaly 5					11.87 [4.95]		3.26 [1.39]
Anomaly 6						17.28 [6.46]	9.55 [3.37]
mkt	-1.72 [-1.27]	-1.20 [-0.86]	-2.60 [-1.84]	-1.01 [-0.70]	-2.63 [-1.87]	-2.47 [-1.77]	0.25 [0.18]
smb	4.67 [2.34]	11.02 [5.34]	8.33 [4.08]	7.25 [3.33]	5.39 [2.58]	6.90 [3.42]	7.46 [3.50]
hml	-8.32 [-3.15]	-7.40 [-2.71]	-9.60 [-3.51]	-9.80 [-3.56]	-9.81 [-3.60]	-11.30 [-4.19]	-5.71 [-2.19]
rmw	-7.50 [-2.82]	0.53 [0.19]	-3.43 [-1.24]	-5.61 [-1.98]	-3.95 [-1.43]	-7.00 [-2.56]	-2.31 [-0.83]
cma	10.59 [2.61]	17.35 [4.33]	21.33 [5.32]	18.18 [4.24]	20.80 [5.18]	21.94 [5.54]	10.40 [2.54]
umd	2.96 [2.15]	3.98 [2.86]	4.56 [3.24]	4.59 [3.18]	3.97 [2.79]	3.31 [2.33]	1.18 [0.86]
# months	732	732	732	630	732	732	630
$\bar{R}^2(\%)$	19	15	12	13	12	14	26

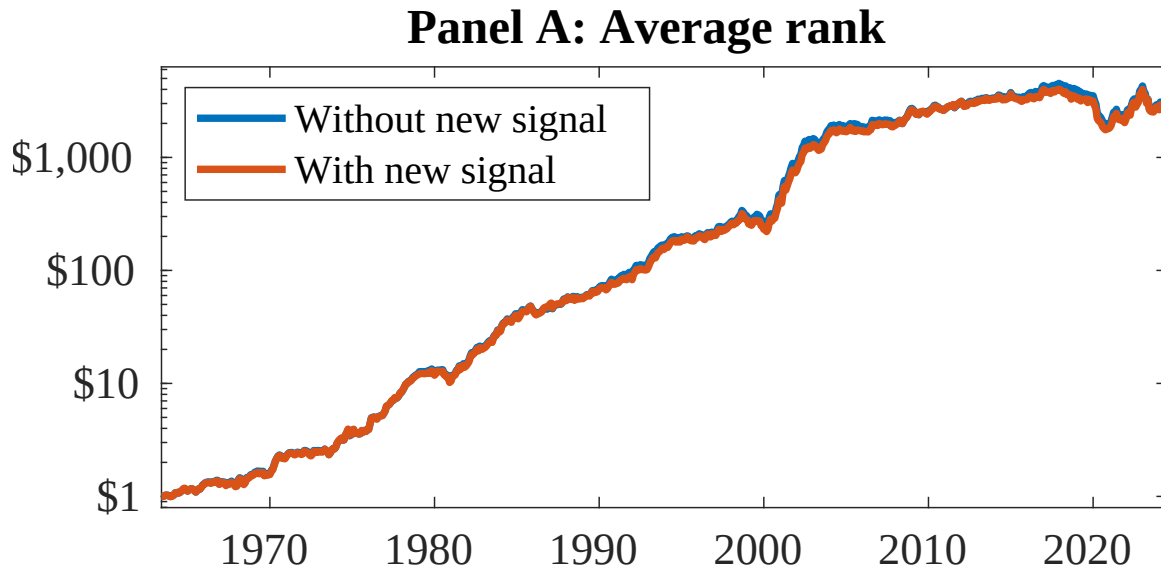


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as DCP. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Bradshaw, M. T. (2011). Analysts’ forecasts: What do we know after decades of work? *Working Paper*.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- DeAngelo, H., DeAngelo, L., and Whited, T. M. (2011). Capital structure dynamics and transitory debt. *Journal of Financial Economics*, 99(2):235–261.
- DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

- Hirshleifer, D. and Teoh, S. H. (2009). The psychological attraction approach to accounting and disclosure policy. *Contemporary Accounting Research*, 26(4):1067–1090.
- Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.
- Lemmon, M. L., Roberts, M. R., and Zender, J. F. (2008). Back to the beginning: Persistence and the cross-section of corporate capital structure. *Journal of Finance*, 63(4):1575–1608.
- Myers, S. C. and Majluf, N. S. (1984). Corporate financing and investment decisions when firms have information that investors do not have. *Journal of Financial Economics*, 13(2):187–221.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Sloan, R. G. (1996). Do stock prices fully reflect information in accruals and cash flows about future earnings? *The Accounting Review*, 71(3):289–315.
- Zhang, X. F. (2006). Information uncertainty and stock returns. *Journal of Finance*, 61(1):105–137.